

Turn/River SDR Agent — Demo Build Specification

Version: 0.1-demo

Execution mode: implementation-support

Demo objective: Demonstrate a governed AI SDR workspace that turns a portfolio-company brief into an approved ICP, researched lead queue, personalized sequence draft, and a staged automation handoff. Nothing is sent externally during the demo.

1. Product definition

Mission

Build an internal multi-tenant SDR Agent platform for Turn/River operating teams and portfolio companies. It should compress the path from GTM context to validated prospects and ready-to-launch outreach, while retaining human approval and an audit trail.

Primary job to be done

When an operating partner or portfolio-company GTM lead needs a repeatable outbound motion, help them define the correct ICP, find and qualify prospects, produce evidence-grounded personalization, create a channel sequence, and stage records for the approved sending system—without relying on manual research, spreadsheet handoffs, or unreviewed sending.

Demo success path

1. User selects DemoCo workspace and opens chat.
2. User says: Build an ICP for our workflow software, research 10 VP Operations prospects, and create a 4-step sequence.
3. Agent returns an ICP card, a prioritized lead table, and lead-research evidence.
4. User opens a sequence draft and edits/approves it.
5. User clicks Stage 10 contacts to sequence.
6. App displays Awaiting approval / mock delivery and writes an audit event—never sends email.

2. Scope and non-goals

In the working demo

- Workspace switcher: Turn/River Internal + DemoCo portfolio company.
- Chat interface with tool-event timeline.
- ICP builder with firmographic, role, pains, triggers, exclusions, and scoring weights.
- Research queue using seeded demo accounts/contacts, with source URLs and confidence labels.
- CSV lead import, dedupe, and field mapping.
- Lead table with New, Researching, Qualified, Review, Staged, Enrolled, Suppressed states.
- Sequence studio that generates a four-step email sequence and personalization snippets.
- Approval drawer that stages enrollment only.
- Dashboard metrics populated from seed data: leads researched, qualified, staged, reply rate, meetings, and pipeline influence.
- Knowledge base file registry and configuration panel.
- Automation Center showing webhook payloads and n8n workflow statuses; actual execution is disabled by default.

Explicitly out of scope for today

- Live email sending, domain warm-up, inbox rotation, or sending-provider credentials.
- Autonomous scraping, LinkedIn automation, or contact-data acquisition without licensed sources and company policy review.
- Production RBAC, SSO, billing, real webhook credentials, or full CRM synchronization.
- Claims of deliverability, response rate, or contact data accuracy based only on generated/demo data.

3. ROSTR operating model

Master agent: TurnRiverSDR

Identity: A governed sales-development orchestration agent for Turn/River and portfolio-company GTM teams.

Objective: Convert a company's approved GTM knowledge into auditable, human-approved ICPs, researched lead lists, outreach assets, and automation handoffs.

Inputs: workspace context; company/product brief; approved knowledge sources; ICP constraints; imported leads; user requests; integration configuration; policy settings.

Outputs: versioned ICP; research dossier; scored leads; sequence draft; enrollment manifest; dashboard metrics; run ledger; approval request.

Hard boundaries:

- Never send external messages, enroll contacts, export contacts, write to a CRM, or activate a workflow without a specific approval event.
- Never present inferred firmographic/person data as fact; label source, timestamp, and confidence.
- Never retain API keys in database tables, chat logs, artifacts, or source control; use environment-secret references only.
- Do not invent a company's value proposition, case study, compliance posture, or metrics. Ask for a brief or use only approved KB evidence.
- Apply suppression/unsubscribe/do-not-contact status before any staging action.
- For any live integration, run a dry-run preview first and require re-approval if scope, contact count, destination, or sequence changes.

Child agents

Agent	Owns	Produces	Can trigger
PAL Intake	Converts chat request into structured run intent	Intent spec, assumptions, questions	RAG-DAL targets
GTM Strategist	Defines ICP and segments	ICP version, scoring rubric, exclusions	Research plan
RAG-DAL Researcher	Researches accounts and people from allowed sources	Lead dossier, citations, confidence	Qualification proposal
Prospect Qualifier	Scores and routes leads	Score, reason codes, lead state	Review queue
Sequence Architect	Creates editable, channel-aware copy	Sequence JSON, variants, merge fields	Approval request
Automation Planner	Selects approved workflow and builds payload	Dry-run plan, webhook payload, event map	Staging request
Compliance Gate	Enforces suppression, consent, policy, and approval rules	Pass/fail and remediation	Blocks unsafe actions
NPAO Orchestrator	Sets phase, urgency, dependencies, and routing	Run plan, queue priority, handoffs	Child dispatch only

Runtime flow

chat intake → PAL intent spec → GTM Strategist → RAG-DAL research ⇄ Prospect Qualifier → Sequence Architect → Compliance Gate → approval request → Automation Planner dry run → staged enrollment → results ingestion → dashboard

Only the NPAO Orchestrator manages run state. Research and writing agents may produce artifacts; they may not perform side effects.

4. Information architecture

Navigation

- **Dashboard:** KPI tiles, active runs, approval queue, funnel, activity feed.
- **Agent Chat:** threaded chat, visible artifacts, tool timeline, suggested tasks.
- **Leads:** import CSV, filters, scores, research dossiers, suppression state.
- **Sequences:** templates, sequence editor, personalization preview, version history.
- **Automations:** connector cards, dry-run logs, webhook events, failed runs.
- **Knowledge:** product briefs, ICPs, case studies, positioning, battlecards, playbooks; source status and retrieval scope.
- **Configuration:** workspace, agent instructions, scoring weights, integrations, approval policy.

Core data model

```
organizations(id, name, type, domain, created_at)
workspaces(id, organization_id, name, policy_json)
users(id, workspace_id, role)
knowledge_documents(id, workspace_id, title, source_uri, text, tags, approved_at)
icp_versions(id, workspace_id, name, definition_json, status, version)
accounts(id, workspace_id, name, domain, industry, size, region, research_json)
contacts(id, workspace_id, account_id, name, title, email, status, suppression_reason)
lead_scores(id, contact_id, icp_version_id, score, reason_codes, confidence)
research_artifacts(id, entity_type, entity_id, content_json, citations_json, confidence)
sequences(id, workspace_id, name, channel, definition_json, status, version)
sequence_enrollments(id, sequence_id, contact_id, status, approval_id, provider_ref)
runs(id, workspace_id, intent_json, phase, status, created_at)
run_artifacts(id, run_id, type, content_json, provenance_json, version)
approvals(id, workspace_id, run_id, action, scope_json, status, approved_by, approved_at)
automation_runs(id, workspace_id, workflow_key, payload_json, mode, status, external_id)
audit_events(id, workspace_id, actor_type, actor_id, action, entity_type, entity_id,
```

5. Tool contracts

Implement Gemini function declarations as application-owned TypeScript handlers. The model selects a function and arguments; the server validates, executes, records an audit event, and returns structured results.

```
const tools = [
  { name: 'create_icp', args: { workspaceId: 'string', brief: 'string', constraints:
  { name: 'search_knowledge', args: { workspaceId: 'string', query: 'string', limit:
  { name: 'research_prospects', args: { workspaceId: 'string', icpVersionId: 'string'
  { name: 'score_leads', args: { workspaceId: 'string', contactIds: 'string[]', icpV
  { name: 'generate_sequence', args: { workspaceId: 'string', icpVersionId: 'string'
  { name: 'preview_enrollment', args: { workspaceId: 'string', sequenceId: 'string',
  { name: 'request_approval', args: { runId: 'string', action: 'string', scope: 'obj
  { name: 'stage_enrollment', args: { approvalId: 'string' } } },
```

```
{ name: 'trigger_n8n_dry_run', args: { workspaceId: 'string', workflowKey: 'string' }
};
```

stage_enrollment must reject unless the approval has status approved, is unexpired, its sequence/contact/provider scope exactly matches the preview, and every contact passes suppression checks.

6. Integration posture

Domain	Demo adapter	Production adapter	Permission rule
LLM orchestration	Gemini via Google AI Studio/API	Gemini server API	Server-side API key only
Knowledge retrieval	seeded documents + uploaded files	approved Drive/Notion/repository connectors	Workspace-scoped retrieval
Lead source	seeded CSV	licensed provider/API selected by Turn/River	No unlicensed collection
CRM	local mock	HubSpot adapter	Read first; write after approval
Outreach	mock-outreach provider	selected sequence provider	Stage/preview, then exact scoped approval
Automation	n8n dry-run endpoint	n8n webhook with signed secret	No real execution in demo
State	local SQLite/mock repository	AWS Postgres/Aurora or DynamoDB based on platform decision	Tenant isolation + audit log

7. n8n workflow blueprint

lead-intake-and-enrich

Trigger: CSV import or CRM record created.

Nodes: Validate schema → dedupe → suppression check → account/lead research adapter → score lead → save results → create review task → dashboard event.

approved-sequence-staging

Trigger: approved enrollment manifest.

Nodes: validate approval scope → re-check suppression → transform provider payload → **demo: return mock provider reference / production: call provider** → record event → notify dashboard.

reply-and-result-sync

Trigger: provider or mailbox webhook.

Nodes: verify signature → match contact → classify reply → pause sequence → create CRM task → update metrics → route positive reply to owner.

8. Demo seed data

Seed one portfolio-company workspace (DemoCo) with:

- Product brief: AI workflow platform for mid-market operations teams.
- ICP: US-based B2B software and services companies, 100–1,000 employees; VP Operations, RevOps, COO; trigger = growth, system consolidation, or hiring operations leaders; exclusions = agencies, competitors, existing customers, suppressed contacts.
- 12 accounts and 18 contacts with 8 qualified, 3 review, 2 suppressed, 5 unqualified.
- One four-step sequence: Day 1 insight, Day 3 pain/relevance, Day 7 proof/objection, Day 12 respectful close.
- Three mock automation runs: one succeeded, one awaiting approval, one blocked by suppression.

9. Acceptance criteria

- A user can complete the demo success path in under 5 minutes.
- Each lead score has a numeric score, reasons, and confidence.
- Each research assertion displays source URL or is clearly marked seed/demo data.
- The sequence preview renders merge-field substitutions for at least three contacts.
- Clicking stage without approval produces a visible block and audit event.
- An approved mock staging action changes only the selected contacts to Staged and returns a mock provider ID.
- Dashboard updates from local events with no page refresh.
- Application has zero live sending paths enabled in the demo environment.

10. Build order for the 1 PM demo

1. Scaffold a Next.js TypeScript app with a left navigation shell and dark enterprise UI.
2. Implement local seed data and repository functions before any external integration.
3. Build dashboard, chat, leads, sequence, and approval panels as a coherent click-through.
4. Add Gemini tool-call orchestration behind `POST /api/chat`; use deterministic mock fallback if no key is configured.
5. Implement `preview_enrollment` and an approval dialog; make `stage_enrollment` a local state transition only.
6. Add Automation Center that displays the n8n payload and a mock run result.
7. Test the 5-minute run with a fresh browser session.

11. Hermes / Google AI Studio build prompt

```
Build a polished, demo-ready full-stack web application called "Turn/River SDR Agent"

Use Next.js 14+, TypeScript, Tailwind CSS, and shadcn/ui-style components. Use a local

The app is an internal multi-tenant AI SDR workspace for Turn/River and portfolio companies.

Pages:
1. Dashboard: KPI cards (researched, qualified, staged, replies, meetings), funnel view
2. Agent Chat: conversational UI. Add suggested prompts. Render structured agent artifacts
3. Leads: filterable table with account/contact, title, score, confidence, status, title
4. Sequences: 4-step sequence editor with versions, merge fields, per-contact preview
5. Automations: n8n workflow cards for intake/enrichment, approved sequence staging,
6. Knowledge: document cards tagged Product, ICP, Proof, Positioning, Playbook. Include
7. Configuration: editable agent configuration, scoring weights, connector cards, and

Seed data: Create DemoCo, an AI workflow platform selling to 100-1,000 employee B2B companies.

User interaction flow: In chat, a request to build an ICP should produce a structured output.

Create a reusable domain model/types for Workspace, ICP, Account, Contact, LeadScore, etc.

Deliver all source files, a README with run instructions, and an .env.example containing
```

12. Demo narration

"Turn/River's operators can spin up a portfolio-company workspace with its approved GTM knowledge. The SDR Agent uses an ICP to research and score prospects, shows the evidence behind qualification, and generates account-aware outreach. It doesn't autonomously send: it creates a precise enrollment manifest, checks suppression and policy, and stages the execution only after approval. The same state model lets us connect the portfolio company's licensed lead source, CRM, outreach tool, and n8n workflows later without changing the operator experience."

13. Decisions after demo

- Confirm AWS persistence choice: Aurora/Postgres versus DynamoDB.
- Select the licensed prospect-data provider and outreach platform per portfolio company.
- Define contact-data policy, lawful basis, suppression source, regional policy, and retention period.
- Identify Turn/River identity/RBAC requirements and tenant data-boundary expectations.
- Map existing n8n workflows and define signed webhook contracts.
- Approve which knowledge systems may be connected and whether retrieval is workspace or organization scoped.